

NFTs AND COPYRIGHT: THE EVOLUTION OF DIGITAL COPYRIGHT PROTECTION?

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OWEN GRANT



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NFTs and Copyright: The Evolution of Digital Copyright Protection?

Owen Grant*

Outstanding LLM Dissertations 2022

Abstract

Non-Fungible Tokens NFTs are digitally scarce, non-exchangeable cryptographic tokens that represent an underlying work such as a picture or video and exist on a blockchain, mostly used to trade in digital art and collectibles. They are the most recent blockchain development and offer a great deal of promise for the future in numerous sectors. Despite this they are unregulated and suffer from a bad reputation and illegitimacy that exists across current public blockchains and cryptocurrency, due to fraudulent actors and misconceptions of what is owned with an NFT.

In this paper I posit that NFTs could be used to upgrade Digital Rights Management (DRM) through two possible solutions; a copyright register on a blockchain or moving DRM to blockchain – Distributed Digital Rights Management (DDRM). The objective of these solutions is to solve the ongoing problem of digital piracy, which DRM has never been successful in stopping through present-day encryption or content-blocking services. The other benefit is for authors to be guaranteed fair remuneration for their works by cutting out some unnecessary intermediaries and issuing licences through smart contracts. I consider that if successful, these solutions constitute an evolution of digital copyright protection, though they must first overcome legal, practical, and logistical problems.

For either solution to be successful, smart contracts must be able to constitute valid legal contracts with binding obligations as all NFTs contain these to execute terms set by the person creating the token (minter). I show that under current Scots law and under England and Wales contract law smart contracts can satisfy formation requirements of a contract, but they still must overcome the challenges presented by blockchain and a lack of intermediaries such as an established dispute resolution mechanism.

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I demonstrate in this paper that current digital copyright protection could be upgraded with the use of NFTs, specifically in giving authors direct control over the following DRM functions for which previously they were beholden to intermediaries: Assignment; Licensing; Royalty payments; and Registration. Despite this I outline that each solution must overcome some significant challenges, leading me to conclude that a blockchain copyright register utilising NFTs is not likely to be attempted in the near future but could offer huge benefits for exploiters as well as authors. DDRM is a solution already being developed by RAIRtech and is by its nature an idea that numerous companies can compete to develop best.

In the final analysis I conclude that despite their issues NFTs *are* the evolution of digital copyright protection. The bar has been set relatively low with current DRM, meaning an improvement would constitute an evolution, which is evidenced by the lack of reliable ownership information, authors' options for revenue, and present-day piracy statistics.

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Chapter 1 – Introduction to Blockchain, Smart Contracts, and NFTs

In 2021 non-fungible tokens (NFTs) burst into public life as the newest revolutionary technology stemming from blockchain with huge potential in the new era of Web 3.0 tech.¹ Their popularity and hype were so far-reaching that Collins' Dictionary named "NFTs" their word of the year in 2021.² NFTs are the culmination of the previous two revolutionary developments in the crypto world; blockchain, and smart contracts. The total of all NFT sales in 2021 was reportedly between \$11.16 billion³ and \$40 billion,⁴ showing the appetite consumers had for digital art and collectibles, and the potential for authors to profit in a new way. To fully understand what an NFT is, one must understand what blockchain and smart contracts are. What will become clear is that these technologies are interlinked, meaning that developments on blockchain and smart contracts have implications for NFTs.

1.1.

In writing this dissertation I followed a doctrinal, black letter methodology, conducting mostly a literature review of pre-legal structures. In developing my research question and area, what became clear very early is that NFTs occupy a unique space in the intersection of property law; intellectual property law; and contract law, involving international law elements of these. This meant that I needed to understand and review the relevant literature in a broad subject before I was able to narrow my focus back to digital copyright protection. The aim of my research was first to understand what NFTs were, identify what legal issues surrounded them, and to identify a legal challenge that this innovation could potentially solve. This led me to identify piracy, then taking a wider view, to identify digital copyright protection through which Digital Rights Management could be addressed.

I chose this methodology mostly out of necessity; the field I was researching is too broad, ambiguous, and is in constant development. The best option I had was to firstly focus on what the law in each area is, after which it became very clear what aspects of NFTs were problematic in a legal sense. Defining what legal challenges they faced was the safest route for me to take, as if I focused too much on technological challenges these could be made redundant over

¹ See Vermaak, W., "What is Web 3.0?", Coin Market Cap 2021 at <<https://coinmarketcap.com/alexandria/article/what-is-web-3-0>> accessed 15/08/2022.

² Collin's Dictionary, "Word of the Year" at <<https://www.collinsdictionary.com/woty>> accessed 15/08/2022.

³ Statista Research Department, "Sales Value of Art and Collectibles NFTs Worldwide 2019-2021", 12th May 2022 at <<https://www.statista.com/statistics/1299636/sales-value-art-and-collectibles-nfts-worldwide/#statisticContainer>> accessed 15/08/2022.

⁴ Chainalysis, "NFT Transaction Activity Stabilising in 2022 After Explosive Growth in 2021", 5th May 2022 at <<https://blog.chainalysis.com/reports/chainalysis-web3-report-preview-nfts/>> accessed 12/08/2022.

the course of my writing, whereas the legal challenges I had identified were not in the process of reform, and the technological solutions to these problems I identified are still a long way from completion. The subject and direction I wanted to take this paper necessitated that I broaden my sources of research and analysis from purely legal sources and study industry experts in technology journals and in blogs.

The biggest limitation of my research was the topic I had chosen; there is very limited legal analysis of NFTs in the areas relevant to me, forcing me to review journals on computer science and artificial intelligence to gain as much insight as possible. NFTs are not currently addressed in any official legislation, meaning that I conducted mostly a literature review backed by pre-existing black letter sources on surrounding areas. Fortunately, there is a wealth of legal analysis of blockchain and smart contracts, meaning I only had to find the connection between this literature and the largely unexplored area of NFTs.

1.2.

Blockchain, or “distributed ledger technology” (DLT), is a decentralised network of computers which communicate with each other (nodes) and where information is stored in blocks that, once full, are locked and verified by the nodes then added to a chronological chain of other blocks with a timestamp.⁵ The timestamp allows the blockchain to serve as a timeline of events as once on a chain, blocks are immutable, meaning they cannot be altered, but the information inside can be viewed by anyone. Parties are identified by their crypto wallet (like a bank account used to trade in crypto currency or other crypto assets) which has its own unique digital address on a blockchain and can hold tokens and cryptocurrency, allowing parties to remain anonymous yet still traceable. The information entered to the chain must first be verified by nodes to ensure that it is correct information. Under a proof-of-work model, nodes verify blocks by completing mathematical equations of increasing difficulty and energy consumption, the first to complete this will receive payment in the form of cryptocurrency (1 Bitcoin for example). This method of information verification allows blockchains to be “trustless” because parties do not need to place their trust in “any one stranger, institution, or third party in order for a network or payment system to function.”⁶ Users can remain anonymous and do not need to trust the other party to hold up their end of the deal, they must only trust that nodes are acting in their own self-interest by being paid to verify blocks. Having no reliance on trust allows blockchain networks to offer

⁵ Hayes, A., “Blockchain Facts: What Is It, How It Works, and How It Can Be Used” Investopedia, Updated 24th June 2022 at <<https://www.investopedia.com/terms/b/blockchain.asp>> accessed 15/08/2022.

⁶ Cryptopedia, “What Does Trustless Mean?” Gemini 28th June 2022 at <<https://www.gemini.com/cryptopedia/trustless-meaning-blockchain-non-custodial-smart-contracts>> accessed 30/08/2022.

users a way to cut out intermediaries (such as banks) and make transactions with anonymous individuals reliably. There are numerous types of blockchain, including those run publicly which are accessible by everyone using a standard web browser, and those run privately, which control who can participate in it, each with their respective benefits. Blockchains will undoubtedly be a feature in future technology, and efforts are already being made to move land registers on chain⁷ because of its compatibility in tracking and storing a timeline of information, which is especially useful for intellectual property in proving who was first to register a creation.

Blockchain's greatest downfall is its energy inefficiency; with each block comes a larger network to uphold and increasing energy prices upload information into blocks, aptly named "gas", leading to major overall energy usage and giving rise to environmental concerns from this expenditure – Ethereum⁸ (ETH), a public blockchain where NFTs are primarily traded, consumes 112 terawatt-hours of electricity per year – comparable to that of the Netherlands.⁹ For the purposes of this paper I am not focusing on climate issues as I am optimistic that mainstream blockchain systems will adopt a proof-of-stake model which will dramatically decrease energy usage, as instead nodes are chosen to validate blocks based on the number of coins they stake.¹⁰ This model also offers more security because more nodes can verify blocks, so economic penalties for attacking the chain through acquiring 51% of its required computing power are much greater.¹¹

1.3.

Smart contracts are open-source automated self-executing contracts that operate on an "if-then" basis set under conditions determined by the creator.¹² The emergence of blockchain and cryptocurrency popularised the term, despite the concept dating back to 1998, when Nick Szabo

⁷ Rizzo P., 'Sweden Tests Blockchain Smart Contracts for Land Registry' (2016) Coinbase at <<https://www.coindesk.com/markets/2016/06/16/sweden-tests-blockchain-smart-contracts-for-land-registry/>> accessed 11/07/2022.

⁸ See <https://ethereum.org/en/>.

⁹ Ethereum, "Ethereum Energy Consumption", Updated 26th August 2022 at <[https://ethereum.org/en/energy-consumption/#:~:text=Digiconomist%20provides%20whole%2Dnetwork%20energy,\(53%20MT%2Fyr\)>](https://ethereum.org/en/energy-consumption/#:~:text=Digiconomist%20provides%20whole%2Dnetwork%20energy,(53%20MT%2Fyr)>)> accessed 12/08/2022.

¹⁰ Frankenfield, J., "Proof-of-Stake", 9th June 2022, Investopedia at <[https://www.investopedia.com/terms/p/proof-stake-pos.asp#:~:text=Proof%20of%20Stake%20\(POS\)%20is,validating%20or%20becoming%20a%20validator](https://www.investopedia.com/terms/p/proof-stake-pos.asp#:~:text=Proof%20of%20Stake%20(POS)%20is,validating%20or%20becoming%20a%20validator)> accessed 12/08/2022.

¹¹ Ethereum, "What Is Proof-Of-Stake?" at <<https://ethereum.org/en/developers/docs/consensus-mechanisms/pos/>> accessed 12/08/2022.

¹² Frankenfield, J., "Smart Contracts" Investopedia, 24th March 2022 at <https://www.investopedia.com/terms/s/smart-contracts.asp> accessed 11/07/2022.

first coined the idea using the example of a vending machine.¹³ The owner of a vending machine has set the terms of the smart contract as “if” the correct payment is entered, “then” dispense the item. There are no further complications, anyone with money can enter the contract and no one is alienated. Smart contracts work very well with blockchain as once these are stored in blocks they cannot be altered and will carry out their contracts without failure – making them ideal for enforcing payment obligations, which is why they are instrumental in the developing decentralised finance (DeFi) industry.

Smart contracts have competing definitions, characterised by their function,¹⁴ which are discussed in Chapter 2. Importantly, anything entered to the blockchain is through utilising a smart contract, therefore all NFTs contain smart contracts.

1.4.

NFTs are unique digitally scarce tokens used to represent any underlying asset capable of being represented digitally,¹⁵ their individuality allows each NFT to be identified separately with a clear trace of ownership. There are numerous types of tokens that can be used to represent shares, commodities, and various other asset types, but the concept of “fungibility” sets NFTs apart. Fungible means exchangeable, for example a £10 note is fungible as it can be replaced with another £10 note, or for two £5 notes. These tokens are not fungible as they cannot be traded or exchanged at equivalency because each NFT is the only version of itself in existence,¹⁶ but this does not stop two parties agreeing their separate NFTs are worth the same.

At its core an NFT is metadata (data that provides information about other data)¹⁷ of its underlying asset, along with other identifying information contained within the smart contract used to write the token into the chain. There are different types of NFTs varying by the information stored within the smart contract, the crucial information is the tokenID, which is generated on creation of the token; and a contract address, which is where it can be found on chain by anyone with a blockchain scanner.¹⁸ The combination of tokenID and blockchain address

¹³ Szabo N., “The Idea of Smart Contracts” (1997) at <https://www.fon.hum.uva.nl/rob/Courses/InformationInSpeech/CDROM/Literature/L0Twinterschool2006/szabo.best.vwh.net/idea.html> accessed 11/07/2022.

¹⁴ See Stark, J. “Making Sense of Blockchain Smart Contracts”, Coindesk, 11th September 2021 at <https://www.coindesk.com/markets/2016/06/04/making-sense-of-blockchain-smart-contracts/> accessed 11/07/2022.

¹⁵ Guadamuz, A. “The Treachery of Images: Non-Fungible Tokens and Copyright” J.I.P.L.P. 2021 Vol. 00, No. 0 page 3.

¹⁶ Sharma, R., “Non-Fungible Token”, 22nd June 2022, Investopedia at <https://www.investopedia.com/non-fungible-tokens-nft-5115211> accessed 25/08/2022.

¹⁷ Merriam Webster, “Metadata” at <https://www.merriam-webster.com/dictionary/metadata> accessed 30/08/2022.

¹⁸ N15 page 4.

is what makes the token unique. Additional information can be added to the contract, such as: the wallet address of the creator; a link to the underlying work either on the internet or an address on a decentralised cloud storage platform – InterPlanetary File Storage.¹⁹ In some cases the underlying asset is stored within the smart contract, however gas prices often limit this. To “mint” (create a record on the blockchain) the NFT, the “minter” uses a digital image to create a unique number that is written onto the chain using a token standard, such as “ERC-721,” in the form of a smart contract. This is signed by the unique crypto wallet of the creator,²⁰ and this is what gives the token its digital scarcity and thus its value.

NFTs are currently auctioned under the premise of digital art and collectibles, with some claiming to transfer copyright ownership through the sale of the NFT. Beeple sold his piece “Everydays: The First 5000 Days” for \$69 million,²¹ and Jack Dorsey sold a signed NFT of his first ever tweet for \$2.9 million,²² to name two of the most famous examples. What should be clear is that an NFT *is not* the underlying asset – it is a separate entity that *represents* the underlying asset, a digital receipt, in a sense. A large issue surrounding NFTs is the fraudulent activity that occurs; anyone can make an NFT of anything capable of being represented digitally and they can make as many NFTs of this asset as they wish – each token is one of a kind, but there can be numerous tokens of the same work. Therefore, owners have proof of their NFT ownership, but this does not prove ownership of an underlying asset.

NFTs are property in England and Wales²³ and would likely be found as such in Scotland, as they have the hallmarks of personal property: the owner can transfer it, give it to someone else, and exclude others from it.²⁴ The purchaser therefore has property rights in the token, no rights in the underlying asset, but the token has an unfalsifiable relationship to the underlying asset.²⁵ This relationship, whilst unfalsifiable, is yet to be proven to be legally valid as there is no official legal tethering, the seller could easily lie to the buyer about the ownership information of the

¹⁹ IPFS, “How IPFS Works” at <<https://ipfs.io/#how>> accessed 25/08/2022.

²⁰ N15 page 4.

²¹ Kastrenakes, J., “Beeple Sold an NFT for \$69 million” The Verge (11th March 2021) at <<https://www.theverge.com/2021/3/11/22325054/beeple-christies-nft-sale-cost-everydays-69-million>> accessed 25/08/2022.

²² Reuters, “Man Who Paid \$2.9m for NFT of Jack Dorsey’s First Tweet Set to Lose Almost \$2.9m” The Guardian 14th April 2022 at <<https://www.theguardian.com/technology/2022/apr/14/twitter-nft-jack-dorsey-sina-estavi>> accessed 25/08/2022.

²³ *Lavinia Deborah Osborne v (1) Persons Unknown (2) Ozone* [2022] EWHC 1021 (Comm).

²⁴ Castor, A., “U.K.’s High Court Ruling on NFTs as “Property” Has Been Called a Landmark – But It May Not Actually Change Much”, Artnet (4th May 2022) at <<https://news.artnet.com/market/uk-high-recognizes-nfts-as-property-2108605#:~:text=As%20far%20as%20the%20internal,tax%20purposes%2C%E2%80%9D%20he%20sa id.>>> accessed 13/07/2022.

²⁵ Fairfield, J.A.T., “Tokenized: The Law of Non-Fungible Tokens and Unique Digital Property” (2022) 97 Ind LJ 1261 at 1280.

underlying asset. For this reason, NFTs should not be tethered; there is vast room for regulation over the origin of tokens and the legitimacy of sellers which has led to widespread copyfraud and other fraudulent listings.²⁶ Adding to this is an argument which states that fraudulent listings do not constitute copyright infringement due to the separation of the token and work. Therefore, the process of minting must be clarified, as this act is legally ambiguous in copyright terms.

Guadamuz contends that restricted acts of copyright are not breached under current interpretation of copyright law as an NFT is technically a string of numbers²⁷ (this was an argument advanced by the defendant in *Soleymani v Nifty Gateway LLC*).²⁸ In theory this argument is convincing, as there could not be reproduction, adaptation, or substantial copying of a work unless the court is prepared to rule that using metadata of a work in tandem with unique numbers generated separately from the work constitutes infringement. If a fraudulent minter were to keep the NFT locally and not include a link to the infringing work then there could not be a communication to the public either, added to the argument that the NFT is just numbers so it could not be an adaptation or reproduction either. Evidently a solution is required, as the spirit of copyright protection is violated because authors theoretically no longer have an economic monopoly over their works.

Despite this, most instances of infringing listings are settled out of court, as the major NFT listing platforms on the Ethereum blockchain (such as OpenSea,²⁹ Mintable,³⁰ and Rarible)³¹ recognise wrongdoing in minting and auctioning another person's work without expressed authorisation, and thus remove fraudulent listings upon issuance of a complaint.³² This loophole is credible yet solvable; there must be an infringing act in the process of minting a work, but if this were legislated to be an act restricted to the copyright owner then UK courts would not have to stretch interpretations of copyright law. Alternatively, the metadata of a work could be protected, as there is no way for an NFT to obtain its unique string of numbers without the input of that specific work. Notably, the author's moral rights attached to the work may not be violated either due to the separation of token and work.

²⁶ Guadamuz, A., "Copyfraud and Copyright Infringement in NFTs", TechnoLlama (14th March 2021) at <<https://www.technollama.co.uk/copyfraud-and-copyright-infringement-in-nfts>> accessed 6/07/2022.

²⁷ See N15 page 12.

²⁸ [2022] EWHC 773 (Comm) at 34.

²⁹ OpenSea at <<https://opensea.io/>> accessed 25/08/2022.

³⁰ Mintable at <<https://mintable.app/>> accessed 25/08/2022.

³¹ Rarible at <<https://rarible.com/>> accessed 25/08/2022.

³² N15 page 11.

1.5.

Leaving these issues aside, I intend to focus on the future direction of NFTs in evolving digital copyright protection as a form of digital rights management (DRM). DRM is the combination of Rights Management Information (RMI)³³ and Technological Protection Measures (TPM),³⁴ a process of controlling access to IP protected works on digital systems through various methods.³⁵ The general purpose of DRM is to stop online IP infringement and ensure that creators are receiving fair remuneration for the use of their works, and to prevent piracy.³⁶ On the surface it appears that NFTs can complement these aims, given their current use which has led to authors being able to have more control over their works and receive payment directly.

DRM can be executed through methods such as encryption, requiring a key to unlock; geo-blocking/regional restriction; specialised hardware; disposable media and many others. DRM does have credible criticism due to its overbearing nature and limitation of protected uses such as fair dealing,³⁷ which is a crucial component to the rationale for giving authors an economic monopoly over their works. It is not my intention to rehash these arguments, I instead take DRM to be a necessary reality but with scope for improvement. The degree of success of DRM is dependent on the type of content it is protecting – for example, online video games can be protected far more easily than a TV show due, to login requirements and the constant interactivity with other players, as opposed to the downloadability of a TV show.

The failure to protect against piracy in TV and movies³⁸ is arguably due to flaws embedded in Web 2.0³⁹ – the current iteration of the internet and predecessor to Web 3.0, characterised by file-sharing among users, content uploads and an ability to interact online. Due to digital content being so easily copied at high quality for almost no expense, the logic of the argument is that this is a problem that current technology cannot solve. Web 3.0 is the next generation;⁴⁰ characterised by blockchain systems, decentralisation, a lack of intermediaries, and smarter

³³ WIPO Copyright Treaty 1996 Article 12.

³⁴ WIPO Copyright Treaty 1996 Article 11-12 and WIPO Performances and Phonograms Treaty Articles 18-19.

³⁵ See Tran, J.L., "A Primer on Digital Rights Management Technologies" in Lemmer, C.A. and Wale, C.P. "Digital Rights Management: The Librarian's Guide" Rowman & Littlefield, 2016.

³⁶ See Lemmer, C.A. and Wale, C.P. "Digital Rights Management: The Librarian's Guide" Rowman & Littlefield, 2016 – page 2.

³⁷ Copyright Designs and Patents Act 1988 s30.

³⁸ Digital TV Research, "Global Online TV and Movie Revenue Lost Through Piracy From 2010 to 2022", October 2017 at <<https://www.statista.com/statistics/778338/global-online-tv-movie-revenue-loss-piracy/>> accessed 25/08/2022.

³⁹ Pech, S., "Copyright Unchained: How Blockchain Technology Can Change the Administration and Distribution of Copyright Protected Works", 18 NW. J. TECH. & INTELL. PROP. 1 page 12.

⁴⁰ N1.

applications.⁴¹ This offers tokenisation through NFTs which can reintroduce scarcity into the digital world,⁴² with the potential to halt piracy.

Pech, a proponent for moving the copyright registry to blockchain, believes this could solve the following DRM and copyright issues: reliable ownership information; fragmentation of ownership; transparency in content usage and payments; inequality in revenue distribution; and piracy.⁴³ In Chapter 3 I outline these issues to show how my two proposed future uses of NFTs – a blockchain copyright registry (with works issued as NFTs); and distributed digital rights management (DDRM)⁴⁴ – may be equipped to deal with these problems in ways that Web 2.0 cannot.

In summary, NFTs offer a great deal of promise for the future if we look beyond its current unregulated, and flawed uses. There is scope for their use as DRM and protection against copyright infringement, but they will not be the final solution to this problem. For this reason, I believe that evaluation of use cases in the form of a state-led blockchain copyright registry, and a less overbearing, consumer and developer-led solution in the form of DDRM, will give accurate insight into realistic uses for NFTs without succumbing to the surrounding hype.

Chapter 2 – Legally binding smart contracts

For any discussion of my two proposals of a blockchain copyright registry and distributed digital rights management to be valid, I must analyse the validity of smart contracts as valid legal contracts. Upon inspection, there are several issues with smart contracts affecting their *suitability* as legal contracts, these relate to more than just smart contracts but also to the nature of blockchains.

First, I must define what I mean by smart contracts for the purposes of this paper as it is easily assumable that smart contracts mean legal contracts on a blockchain, which is not technically what smart contracts *are*, but in some cases, what they *do*. As I set out in the previous chapter, anything that is written into a blockchain is done so through the use of a smart contract, but what they are specifically is automated agreements – this is the definition I prefer, which is referred

⁴¹ N1.

⁴² Bodó B., Gervais D. and Quintais J.P., “Blockchain and Smart Contracts: The Missing Link in Copyright Licensing?” (2018) 26:4 International Journal of Law and Information Technology page 315.

⁴³ N39 page 2.

⁴⁴ Sonnenfeld, G., “Distributed Digital Rights Management: True Autonomy for the Creator” Medium, 9th September 2021 at <<https://rairtech.medium.com/distributed-digital-rights-management-true-autonomy-for-the-creator-88c03b11290>> accessed 29/08/2022.

to as *smart contract code*.⁴⁵ With regard to that which lawyers tend to refer to as the technology being a supplement to or replacement for traditional legal contracts, these are called *smart legal contracts*,⁴⁶ and this is what I will primarily be assessing here. A final definition comes from developers who see smart contracts being able to transcend typical legal contracts, creating unique ways to transact and therefore are not applicable to traditional contracts – these are referred to as *smart alternative contracts*.⁴⁷

These are the core definitions used across various sectors discussing smart contracts, but without delving too deeply into a discussion of definitions, using the vending machine as the first example of a smart contract, there is an inherent legal element as ultimately it is an automated contract of sale – there is an agreed upon price by which a transfer of property takes place.⁴⁸ Szabo's article on smart contracts predates blockchain technology by approximately 10 years so it cannot be the case that smart contracts *mean* "anything that is put on the blockchain." More to the point, smart contracts were at least somewhat intended to automate legal agreements. The question is, are they suitable to facilitate a transfer of copyright on a blockchain network?

The answer to this question has as much to do with the actualities of blockchain as it does with smart contracts themselves; there are several issues with blockchain networks that I will highlight in Part 2.4 but first I will analyse the legal standing of smart contracts.

2.1.

The first body with legislative power to provide a definition of smart contracts is Arizona in the USA, who found:

"Smart contract" means an event-driven program, with state, that runs on a distributed, decentralised, shared and replicated ledger and that can take custody over and instruct transfer of assets on that ledger.⁴⁹

This definition was used as a template in the state legislature of Tennessee.⁵⁰ Detailed analysis and breakdown of the three component parts of the Arizona definition⁵¹ is beyond the scope of this piece, but importantly, this definition excludes smart contracts that do not exist on DLTs,

⁴⁵ Stark, J., "Making Sense of Blockchain Smart Contracts", Coindesk, 11th September 2021 at <<https://www.coindesk.com/markets/2016/06/04/making-sense-of-blockchain-smart-contracts/>> accessed 11/07/2022.

⁴⁶ Ibid.

⁴⁷ Ibid.

⁴⁸ Sale of Goods Act 1979 s2(1).

⁴⁹ Arizona House Bill 2417 (2017) Article 5(E)(2).

⁵⁰ TN Code ss 47-10-201 (2019).

⁵¹ (1) The "event driven programme" requirement, (2) the requirement to run on a distributed, decentralised, shared and replicated ledger, and (3) the requirement to be able to take custody over and instruct the transfer of assets.

which leaves a lot to be desired on a technical level but this also shows the extent to which smart contracts and blockchain appear inextricably linked in all current uses of either technology. The definition of smart contracts used by the UK Law Commission, however, is much simpler and more accurate; “computer programs which run automatically, in whole or in part, without the need for human intervention.”⁵²

2.2.

Can smart legal contracts be valid legally binding contracts? In Scotland a contract is formed when the parties have reached an agreement – *consensus in idem* – with the intention to create legal relations.⁵³ This is an objective test, determined by what a reasonable bystander would infer from judging the actions of the parties, even in the case that subjectively no agreement was concluded.⁵⁴ Contract law is generally open-minded to the acute realities of human relationships and thus has found valid contracts to take many forms,⁵⁵ provided they satisfy formation requirements; aiming to

Facilitate the transactions of commercial men, and not to create obstacles in the way of solving practical problems arising out of the circumstances confronting them, or to expose them to unnecessary pitfalls.⁵⁶

Therefore, the form of a smart contract will not prevent the formation of a binding contract in principle.

There must be an offer and an acceptance; crucially, the offer must contemplate binding obligations being formed as soon as it is met by unqualified acceptance.⁵⁷ Acceptance does not have to be expressed, it can be inferred from the positive actions of the offeree.⁵⁸ Going back to the simple vending machine, there is no Scottish authority in accordance to this, however England and Wales jurisprudence believes this would constitute a standing offer,⁵⁹ meaning that entering money is acceptance of the contract and the machine’s refusal of money would be a breach of contract.⁶⁰ Alternatively the vending machine could be an invitation to treat, the

⁵² Law Commission, “Smart Contracts,” 25th November 2021 at <<https://www.lawcom.gov.uk/project/smart-contracts/#:~:text=Smart%20contracts%20can%20perform%20transactions,of%20a%20legally%20binding%20contract.>> accessed 25/08/2022.

⁵³ MacQueen, H., “Contract Law in Scotland” 4th Edition (Bloomsbury Professional, 2016), Page 41 at 2.2

⁵⁴ *Muirhead and Turnbull v Dickson* (1905) 7 F 686.

⁵⁵ See *Carlyle v Royal Bank of Scotland* [2015] UKSC 13 where a phone call was held as a legally binding contract.

⁵⁶ *R&J Dempster Ltd v Motherwell Bridge and Engineering Ltd* (1964) SC 308 at 332.

⁵⁷ *William Lippe Architects Ltd v Innes* [2006] CSOH 182A.

⁵⁸ N53 page 54 at 2.23.

⁵⁹ *Thornton v Shoe Lane Parking* [1971] 2 QB 163.

⁶⁰ N53 page 50 at 2.15.

essential point is that this early example of a smart contract has already satisfied the conditions of offer and acceptance, therefore other “if-then” contracts would also satisfy this requirement.

There is no requirement for a contract to be in writing provided that an agreement is reached with the intention to create legal relations,⁶¹ however assignments and licences relating to copyright works must be signed and in writing.⁶² Where there is a requirement of writing, electronic signatures have been accepted in contracts relating to real rights in land.⁶³ Therefore, the only question to determine is if smart contracts can satisfy the requirement of being “in writing” to be a valid legal contract in cases of assigning copyright. On this point, guidance from the Law Commission’s report on smart legal contracts⁶⁴ is helpful: concluding that in England and Wales the legal framework can facilitate and support the use of smart legal contracts.⁶⁵ Their report mentioned the various forms a smart contract can take: (1) written and agreed in natural language; (2) fully coded; or (3) a hybrid.⁶⁶ Their main findings⁶⁷ were complications in finding an intention to create legal relations if the agreement was not first made clear in natural language,⁶⁸ and the requirement of “in writing” being satisfied if the terms of a contract are in a lower level of code than “source code.”⁶⁹

In all cases there must be predetermined dispute resolution clauses, much like traditional paper contracts: setting out which courts have jurisdiction and in what circumstances. Smart legal contracts should not omit clauses that are found in their offline counterparts.

In summary, the formation requirements of a contract are theoretically able to be satisfied by smart contracts, provided that the terms of a contract are agreed upon in natural language. This means that if two parties agree terms of a contract and set them out in the smart contract of an NFT, after which the NFT is sold – there would be a binding legal contract.

2.3.

There are some issues that must be raised beyond the format of a smart contract which relate to their *suitability* in replacing a traditional legal contract. The main concern with the current use of NFTs in selling digital art in an unregulated environment is the anonymity of parties. There is no legal requirement to know the identity of the other party in a contract of sale (evidenced by

⁶¹ Requirements of Writing (Scotland) Act 1995 s1 (RWSA).

⁶² Copyright Designs and Patents Act 1988 s90(3).

⁶³ RWSA 1995 ss9A-9G.

⁶⁴ N52

⁶⁵ Ibid.

⁶⁶ Law Commission, “Smart Legal Contracts: Summary” LC 401, 25th November 2021 page 8.

⁶⁷ See Law Commission *supra* note 66.

⁶⁸ N66 page 12.

⁶⁹ N66 page 13.

each party's anonymity in a vending machine transaction), however, the anonymity of parties on DLTs is still a concern. For a contract to be valid the parties must have capacity to enter contracts, meaning children and those mentally impaired are protected.⁷⁰ Therefore, a child with sufficient technical knowledge could buy an NFT because Ethereum is a public blockchain with no entry requirements to participate.⁷¹ This is not a problem unique to NFTs however, users can purchase items online whilst drunk or at any online retailer – the solution to this is provided by a statutory 14-day refund for all goods sold online, by mail, or over the phone.⁷² This could possibly apply to NFTs, although their sales are more akin to auctions where the listing is an invitation to treat and offers are made by bidders.⁷³ The new Ethereum token standard, ERC-721R, adds an option of a 14-day refund to the NFT⁷⁴ which could be a solution if the seller is deemed to be the offeror rather than the offeree. Clearly, this is another area that requires clarification, but it does not affect the validity of smart legal contracts.

One solution to the problem of minors purchasing NFTs is to impose age restrictions or parental oversight on the entrants to blockchains or to opening crypto wallets. This could be altered in cases of minors trading in their own IP, which may include a countersignature from a parent or guardian's crypto wallet. A provision along these lines is likely to be used in the case of any public registry that is moved to blockchain given the need to authenticate and verify that all entries to the chain are legitimate.

2.4.

Although they can satisfy all formation requirements, one issue with smart legal contracts on blockchains is the potential for security threats putting the entire blockchain at risk. Attacks on blockchain can come in the form of "51% attacks" whereby miners can collaborate their resources to gain control of 51% of the computing power used to uphold the blockchain, which then enables them to stop the recording of new blocks and reverse transactions made while they are in control.⁷⁵ This is difficult on large scale blockchains such as Ethereum, due to energy

⁷⁰ Sale of Goods Act 1979 s3.

⁷¹ See Ethereum, "Ethereum Accounts", Last Edit: @cortwines, 3rd January 2022 at <<https://ethereum.org/en/developers/docs/accounts/#account-creation>> accessed 24/08/2022, and Parentzone, "Cryptocurrency" at <<https://parentzone.org.uk/article/cryptocurrency#:~:text=How%20old%20do%20you%20have,age%20can%20mine%20for%20cryptocurrency.>> accessed 24/08/2022.

⁷² Gov.UK, "Online and Distance Selling" at <<https://www.gov.uk/online-and-distance-selling-for-businesses/online-selling>> accessed 30/08/2022.

⁷³ Contract Law in Scotland page 51 at 2.16.

⁷⁴ Fernau, O., "Refunds for NFTs? New Standard Sets Give-Back Period for Minters" The Defiant, 12th April 2022 at <<https://thedefiant.io/nfts-refunds-standard-erc721r>> accessed 29/08/2022.

⁷⁵ Frankenfield, J., "51% Attack", Investopedia, 14th July 2022 at <<https://www.investopedia.com/terms/1/51->

costs, but smaller ones are more easily targeted in this way. Security threats of this sort must be protected against for any official registry through proof-of-stake modelling, or government sanctioned servers or nodes to create high barriers to entry. This is less of a credible threat but a concern that must be borne in mind with using smart legal contracts.

The immutability of smart contracts does not lend itself well to binding legal contracts – as a bug in the code can lead to a user being locked out of the contract forever. Whilst traditional contracts are similarly locked once entered, they are not subject to the same frailties that a simple glitch in the code can have in affecting the whole running of a contract – an error akin to a simple “typo” could have far greater consequences. This happened to the developers of Akutars NFTs, where a bug in their smart contract left \$34 million worth of ETH locked away forever.⁷⁶ Bugs in smart contracts are not rare either, one study of 19,366 smart contracts found that 44% of them contained bugs,⁷⁷ another found 25% of smart contracts they reviewed contained bugs in the code.⁷⁸ This will lead to users paying for their smart contracts to be coded or proof-checked, exchanging traditional legal fees with these new fees, which are likely to increase as reliance on blockchain becomes greater.

Even with all the current issues with smart contracts today, there is no reason to believe these cannot be improved upon. Smart contracts are still a recent development with a small pool of experienced developers,⁷⁹ this will change in time. Scholars have given smart contracts attention for their prospective uses in various sectors such as dispute resolution,⁸⁰ the automation of remuneration,⁸¹ and importantly, DRM.⁸² Efforts are being made to improve smart contracts through development of new coding languages, especially in the decentralised finance (DeFi) sector.⁸³ The process of improvement could be hastened by using a trusted

[attack.asp#:~:text=A%2051%25%20attack%20is%20an%20attack%20on%20a%20blockchain%20by,other%20miners%20from%20completing%20blocks.>](#) accessed 26/08/2022.

⁷⁶ Potnis, N., “Flaw in the Smart Contract for Akutars NFTs Locked Up \$34 Million Worth of ETH”, LinkedIn, 26th April 2022 <https://www.linkedin.com/pulse/flaw-smart-contract-akutars-nfts-locked-up-34-million-nameet-potnis?trk=articles_directory> accessed 12/07/2022.

⁷⁷ Olickel H., ‘Why Smart Contracts Fail: Undiscovered bugs and what we can do about them’ (2016) Medium, <https://medium.com/@hrishiolickel/why-smart-contracts-fail-undiscovered-bugs-and-whatwe-can-do-about-them-119aa2843007> accessed 12/07/2022.

⁷⁸ Sedgwick K, ‘25% of All Smart Contracts Contain Critical Bugs’ (2018) Bitcoin News, <https://news.bitcoin.com/25-of-all-smart-contracts-contain-critical-bugs/> accessed 12/07/2022.

⁷⁹ Hine, M., “The Problem with Smart Contracts Today” Radix, 4th October 2021 at <<https://www.radixdlt.com/post/the-problem-with-smart-contracts-today>> accessed 12/07/2022.

⁸⁰ See Koulu R., ‘Blockchains and Online Dispute Resolution: Smart Contracts as an Alternative to Enforcement’ (2016) 13 SCRIPTed 40.

⁸¹ Bodó, B., Gervais, D., and Quintais, J.P., ‘Blockchain and Smart Contracts: The Missing Link in Copyright Licensing?’ (2018) 26:4 International Journal of Law and Information Technology page 19.

⁸² Ibid page 18.

⁸³ N79.

intermediary with oversight of the blockchain who could resolve issues and proof-check entries for bugs.

Looking specifically at copyright functions, smart contracts could be used to automate direct licensing if used in conjunction with a blockchain copyright registry. This is proven by the current standardisation provided in Creative Commons licenses, available on an “as-is” basis,⁸⁴ and already finding a balance between machine readable and natural language licensing.⁸⁵ Therefore, large blockchain repositories of RMI could efficiently manage and license works⁸⁶ provided they also track the usage of works on their chain through methods such as digital fingerprinting, so they can address any issues and tailor their licences accordingly.

DRM is already a largely automated process with two-factor authentication systems, password entries to view content and similar methods of verifying a user is who they claim to be. Smart contract code is already used for decentralised applications (dApps) which is why I believe that DRM can be used on a blockchain using smart contracts – but proving legitimacy would be my main concern if there were no parties monitoring who was using the service.

In summary, smart contracts can satisfy contract formation requirements and be valid legally binding contracts under current UK and Scots contract law, although there are challenges which could present themselves if there is limited oversight of participants and entries. Therefore, a trusted intermediary, whilst not needed for the formation of binding contracts, would be useful in ensuring legitimacy and preventing parties causing themselves financial harm, as seen with Akutars NFTs. Given that smart contracts can be used to create binding legal obligations and have prospective uses in automating licences similar to CC licensing, the next step is to outline and show why my two proposed uses could evolve online copyright protection and DRM.

Chapter 3 – NFTs as a solution to piracy

As established in previous chapters, NFTs can facilitate contractual arrangements such as a transfer/assignment of copyright provided they are intended to create legal obligations and have been expressly agreed, preferably in natural language. This leaves me with the core issues that must be solved and my two proposed solutions. I am using Pech’s model of a blockchain copyright register⁸⁷ and RAIRech’s model of “Distributed Digital Rights Management”⁸⁸ as a

⁸⁴ Creative Commons, “Attribution 4.0 International” at <<https://creativecommons.org/licenses/by/4.0/legalcode>> accessed 26/08/2022.

⁸⁵ Creative Commons, “Three “Layers” of Licenses” at <https://creativecommons.org/licenses/> accessed 26/08/2022.

⁸⁶ N81 page 20.

⁸⁷ N39 Pech.

⁸⁸ N44.

starting point for each solution but these are by no means complete and without their own problems. Their respective creators had acutely different but largely similar problems they set out to address, the difference between them being the likelihood of being used and their probability of success, although their benefits and usage are very similar.

3.1.

The specific issues I look for NFTs to address are inadequate and inefficient DRM, protection against piracy, and *intermediaries abusing their power*.

An example of the inadequacy of DRM is Spotify who are very frequently involved in legal battles over using works for which they do not have licences.⁸⁹ The reason for this is not always because they are acting dishonestly, but because there is very little reliable ownership information storage and access. One study found that the average time it took to track down the current owner of a work and acquire a licence was 18 months,⁹⁰ this is because transfers of ownership, whilst required to be signed in writing, are not recorded on the copyright register. Exploiters are then left with a choice of expending the resources for a long and potentially fruitless endeavour, or to use the work without permission – I believe that consumers should be assumed to act rationally, even if this leads them to make unethical decisions. This is a lose-lose situation for both exploiters and authors; exploiters such as Spotify have the intention to pay for their use where they can, authors similarly lose in this situation as they are losing revenue for use of their works by exploiters who may be willing to pay for their use.

Looking at DRM specifically, one key component is RMI, which is increasingly fragmented as authors assign various agencies different individual restricted economic rights to their works. Publishers control some rights, performance rights are controlled by agencies such as the American Society of Composers, and reproduction rights are controlled by other agencies, such as The Harry Fox Agency. This leads to authors having to pay several agencies for their works to be exploited, and they are increasingly becoming pressured into signing off rights, as there is currently no other option for their works to gain similar revenues without the use of intermediaries.⁹¹ To solve this problem smart contracts have been proven to constitute valid legal contracts, so NFTs could be used to carry out individual DRM functions, namely: assignment; licensing; registration; and royalty payments.

⁸⁹ See Titlow, J.P., "Why Can't Spotify Stop Getting Sued? It's More Complex Than It Sounds, FAST CO. (July 25, 2017) at <<https://www.fastcompany.com/40441194/why-does-spotify-keep-getting-sued>> accessed 30/08/2022.

⁹⁰ Touve, D., "The Innovation Paradox: How Licensing and Copyright Impacts Digital Music Startups" 6 (2012), found in Pech N39.

⁹¹ N44.

Piracy is without a doubt the biggest problem facing digital copyright, with evidence showing that piracy is increasing with each year.⁹² This problem is caused by numerous issues, all coinciding to make the typical consumer more inclined to pirate works. As with the fragmentation of restricted acts of copyright, so too are exclusive works being offered in increasing numbers of paid subscription services which the average consumer cannot afford. Another contributor to piracy is the ease at which high quality direct copies can be made and the almost negligible cost for doing so, meaning there are increasing numbers of works available to view illegally. To solve this problem copyrighted works need to be watermarked and have their usage tracked so that pirates can be tracked down and held accountable. This requires an upgrade to DRM protection measures of the sort made available through Web 3.0 encryption.

Intermediaries such as the agencies mentioned above, as well as internet service providers (ISPs) are the root cause of several issues plaguing creators: little transparency in content usage and revenues; unfair costs of entry; and unequal shares in revenues. This is not news to those involved in cryptocurrency as a removal of all intermediaries is one of the main objectives, and why NFTs may be best placed to deal with these issues as they have no reliance on traditional networks meaning they are not beholden to the same intermediaries. Authors often receive no statistics on the use of their works and thus they do not know how much they are owed and cannot work it out for themselves.⁹³ To add to this, the revenues generated by the music industry increase with each year, thanks to streaming services such as Spotify, yet in 2017 authors only received 12% of these revenues.⁹⁴ The figures themselves do not necessarily show there is injustice, but the lack of information mentioned earlier does create unnecessary uncertainty, and the number of intermediaries receiving a share in revenues should not be the cost of doing business as an author.

There are more issues that can be solved other than those above, and each writer sets these out in their respective articles⁹⁵ but for the purpose of showing how NFTs could revolutionise copyright protection the three I have outlined are most important.

⁹² Spajic, D.J., "Piracy Is Back: Piracy Statistics for 2022" DataProt, Updated 17th May 2022 at <<https://dataprot.net/statistics/piracy-statistics/#:~:text=126.7%20billion%20viewings%20worth%20of,billion%20in%20the%20movie%20in%20industry.>> accessed 30/08/2022.

⁹³ N39.

⁹⁴ Bazinet J.B. et al, "Putting The Band Back Together: Remastering The World Of Music" (CitiGPS, August, 2018), page 63 at <<https://www.citivelocity.com/citigps/music-industry/>> accessed 30/08/2022.

⁹⁵ N39 page 2-8, RAIR Tech.

3.2.

Pech was writing in 2020, a year before NFTs' surge in popularity, but his idea for a copyright register is bolstered, rather than made redundant, in my view. NFTs can be added to a blockchain just as any other tokens are entered on chain, but with the added value of complete digital scarcity, which would increase the efficiency of the system's verification processes across the board. I have thus added the use of NFTs to this model.

How would a blockchain copyright register work? In simple terms, copyrighted works could be entered as NFTs to a blockchain, with all their respective ownership information that can be stored in the smart contract, on an IPFS platform, or in the metadata, and a timestamp of the exact date and time of the transaction. Any transfers of title or restricted rights would be added automatically through the underlying smart contract. If this transaction took place offline a separate entry linking the new information to the initial work could be made. Transfers of individual rights in a work could be issued as separate NFTs or other tokens.⁹⁶

Any additional information, such as the author's date of death, could be added where relevant so that all works have information on their term of protection. A large benefit of using a blockchain register is that any new information can be linked with a prior work and serve as a chain of title.⁹⁷

If authors wanted to use this register for physical works such as a painting, these could have a radio frequency identification (RFID) chip attached. Works using this would require frequent updates to avoid outdated information as new information is put on the register.⁹⁸ For this to work it would need to be a private blockchain, with oversight by a trusted intermediary, such as the Intellectual Property Office, to verify that all parties are legitimate and to serve as an oracle verifying physical entries. This sort of oversight is not problematic in the sense that all intermediaries are not cut out because there must be a stamp of authority. There also must be a solution to the "oracle problem"⁹⁹ of verifying that physical objects have been transferred; smart contracts cannot see the physical world, so they need to rely on a third party of some sort, so it makes sense for this to be the IPO.

3.3.

How this could help tackle piracy is through usage tracking that could be implemented to monitor works on the register. YouTube already uses a content use tracker with its Content ID scanner in

⁹⁶ N39 page 14.

⁹⁷ N39 page 12.

⁹⁸ N39 page 14.

⁹⁹ Guadamuz, A., "All Watched Over by Machines of Loving Grace: A Critical Look at Smart Contracts" *Computer Law and Security Review* 35:6 (2019) page 16.

which creators upload a sample of their work which is then scanned for uses across YouTube.¹⁰⁰ Creators then have the option to monetise or block any uses and track any viewership statistics available.¹⁰¹ This could be improved upon in this proposed use, as works are already uploaded to the chain so there would be no need for creators to request a scan as it could be done automatically. If individual copies of works are issued as tokens themselves then uses online can be verified as legal/illegal, with takedown notices served automatically if the use is beyond the scope of any licence given.¹⁰² Digital fingerprinting could also be used in specific copies issued, so that any illegal use could be tracked to an individual copy and thus a potential culprit for the pirated use. This could deter parties involved in the making of infringing copies because there would now be a credible threat of facing consequences due to this new way to track the origin of each copy to a transaction stored on the blockchain.

RMI is potentially improved by this solution as the current registries maintained by Collective Management Organisations (CMOs) storing information on orphan works and works in the public domain could be stored in the same place. Any legal requirements for uses of works such as diligence searches for owners of orphan works could be encoded on chain as evidence that the necessary search was made, so any use thereafter is lawful.¹⁰³

This also provides a solution to overbearing intermediaries by allowing the rightsholders to essentially be the intermediaries themselves, as they are in control of all aspects of their works. Authors can have a much closer relationship with their fans because they sell to them directly, this also opens alternative revenue streams if, for example, they wanted to sell individual rights to a song as tokens.

However, this solution has some significant issues that must be overcome for it to come to fruition, the biggest of which comes from tokens being legally tethered to the works that they are used to represent. For this register to work as intended, the law needs to recognise that ownership of an NFT attached to a work on the copyright register has a legitimate tethering effect that embodies property rights in the underlying asset. This is compared with other examples of tokens; negotiable instruments; or deeds to real property,¹⁰⁴ which a court will recognise as having a legal connection to the underlying asset. At present there should not be legal tethering due to the illegitimacy surrounding NFTs, but this could be rectified due to the

¹⁰⁰ N39 page 15.

¹⁰¹ Ibid.

¹⁰² Ibid.

¹⁰³ N81 page 17.

¹⁰⁴ Moringiello, J.M. and Odinet, C.K., "The Property Law of Tokens" 1st November 2021 (Florida Law Review, Forthcoming 2022), U Iowa Legal Studies Research Paper No. 2021-44, Widener Law Commonwealth Research Paper, Available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3928901, page 36.

presence of the IPO as a barrier to entry. Without legal recognition of tethering effects, I do not see a copyright register of this sort being successful.

Compliance with the Berne Convention Article 5(2)¹⁰⁵ is a smaller concern, as this prevents the enjoyment of protected rights being subject to any formalities – such as mandatory registration to a blockchain copyright register. A register such as Pech’s does not have to be comprehensive of all works, it rightfully should be left to authors to voluntarily put their works there. There is no official copyright register in the UK presently; if authors want their works protected to a fuller extent, then they will want to register them, similarly they will not be able to complain if they have not registered. The US already has a copyright register with added benefits for registering works, such as the ability to file for civil action for copyright infringement,¹⁰⁶ meaning Berne’s prohibition of formalities must only apply to non-domestic works.¹⁰⁷ If there was an appetite for such a requirement; countries could domestically mandate registration to blockchain registers, and they could be linked internationally.

Another issue is the practicality: the scale of the project; its cost and source of funding; the potential political pressure from established intermediaries likely to lose their revenues; the time it would take to complete; the abundance of false or unreliable ownership information; the scalability and other logistical issues.¹⁰⁸

There is no doubt that on the surface this does not look likely to happen because of the number of practical concerns affecting it from start to finish and the running of it thereafter. Despite this I believe that efforts being made to move the land register to blockchain¹⁰⁹ will serve as a proof-of-concept and inspire other countries to move various registers to blockchain networks, it is just a matter of time.

3.4.

The alternative solution I believe is credible is RAIRech’s DDRM idea, which is an upgraded form of current NFT usage. RAIRech is developing a distributed marketplace where creators can control the terms of the use of their content and monetise their works directly using NFTs. This is not an endorsement of RAIRech themselves, but the model they have developed is valuable as a proof-of-concept. There is limited information on how specifically this will work as it is still under

¹⁰⁵ The Berne Convention for the Protection of Literary and Artistic Works 1866.

¹⁰⁶ N39 page 33.

¹⁰⁷ Ibid.

¹⁰⁸ Ibid 16–35.

¹⁰⁹ Rizzo P, ‘Sweden Tests Blockchain Smart Contracts for Land Registry’ (2016) Coinbase at <<https://www.coindesk.com/markets/2016/06/16/sweden-tests-blockchain-smart-contracts-for-land-registry/>> accessed 11/07/2022.

development. Their idea is similar to what NFTs are already used for, except RAIR has developed a unique cryptographic key system which locks content until it is paid for (should the creator wish it to be locked) – as opposed to other marketplaces where the images are freely viewable and thus open to being copied like any other digital media.¹¹⁰

Therefore, works such as videos or music can be accessed through a form of encrypted streaming, selling their content directly to consumers and thus are not beholden to intermediaries. Authors are also given the ability to select their marketplace they sell to; and are guaranteed royalty payments.

This solution is far simpler than a blockchain copyright register and can be optimised to carry out much of the same functions: transfers of rights can be recorded on chain; usage is trackable in the same way; RMI could potentially be attached; and authors have new ways of monetising their works.

The difference, however, comes in the lack of intermediaries, thus ensuring complete decentralisation, but similarly locking the door on physical goods being traded in this way due to the oracle problem mentioned earlier. There are numerous issues which hamper this solution's chances of success, discussed subsequently, but the main benefits are the scale of the project compared with the blockchain register and the fact that it is a consumer-led idea already being developed, free from the political challenges an adoption of a blockchain copyright register could face.

What is clear upon inspection of both solutions is that if successful they would usurp the need for some aspects of current DRM: all RMI can be put on one copyright register, and DDRM would prevent the need for TPM as the encryption and usage tracking provided covers this function. This is not surprising, as blockchain was developed to provide an alternative to traditional intermediary oversight. One function of many smart contracts in current NFTs is ensuring authors receive fair remuneration for use of their works, added to this are royalty payments that give authors a portion of any future sales of that work, such as the model used with the DADA.NYC platform; where authors receive 30%, platforms receive 10% and the owner/transferor receives the remaining 60%.¹¹¹ This could be implemented with any use of NFTs, and authors can be sure they are being paid what they are owed.

¹¹⁰ N44.

¹¹¹ Evans, T.M., "Cryptokitties, Cryptography, and Copyright" (2019) 47 AIPLA QJ 219, page 265.

The problems that face DDRM are less significant than those facing a blockchain register but they still require addressing. There needs to be clear dispute resolution available for faults in smart contracts, given the wealth of bugs seen in current usage,¹¹² the complete abandonment of intermediaries is in my view a bad idea, but I also do not envisage it preventing DDRM from working, due to ongoing examples of cryptocurrency that operate without intermediaries, such as Bitcoin and ETH.

This idea also suffers from the legal tethering in cases of assignment through NFTs using blockchain – there can be a valid binding contract, but the lack of official recognition is a concern which may stop its use. To help solve this problem I think RAIR need to play a role as an intermediary in verifying network participants and authorship, this does not have to be an overbearing role, but some form of verification would help show legitimacy in the works being exploited on the network. The issue of legitimacy in NFTs could be credited to the fact that Ethereum is a public blockchain where most transactions take place, where everything and anything goes, provided auctioning platforms do not adjudicate. Therefore, if a private blockchain is used, with RAIR gating and vetting entrants, authors could still enjoy digital copyright protection, in an unofficial manner.

Overall, DDRM is a more practical use of NFTs and has a higher probability of being successful. There is not as great a difference between DDRM, and the current digital art/collectibles use showing that the technology can perform the tasks required of it. The key difference is in encryption, which adds a layer of protection to works in a manner that is consistent with traditional DRM.

3.5.

In summary, the two possible uses of NFTs set out above show a capability of evolving digital copyright protection by giving authors the option of controlling their own works to a much greater extent than is currently being practised. There are practical and logistical issues with each, but none of these is fatal to the future implementation of either. The final step is to discuss the remaining concerns I have with NFTs, which could prevent them being adopted and used in the mainstream.

Chapter 4 – Problems preventing their use

So far, I have shown that NFTs can facilitate an assignment or transfer of copyrights in theory, which allows Pech and RAIR Tech's proposals for their use as DRM to prevent piracy to be

¹¹² See Chapter 2.4.

considered seriously. The last step to determine whether NFTs are the future of digital copyright protection is to discuss the final outstanding issues which could either prevent their longevity of use or prevent their use outright.

4.1.

The current lack of regulation is undoubtedly the main concern I have that affects NFTs being used in mainstream digital copyright protection. Until there is clarification over the process of minting and the establishment of ownership of the underlying asset, NFTs will not be a reliable source of copyright protection. The tethering to an underlying asset is less important in the case of DDRM because of the function that this serves, however this would face a different issue of legitimacy. Until there is regulation there can be limited legitimacy and authors will still be beholden to intermediaries to secure their rights.

The Law Commission of England and Wales states that it has every intention of making the UK competitive in digital assets and has published a consultation paper including provisional law reform proposals, although not specifically for NFTs.¹¹³ This could put these worries out of sight as the Commission intends to provide reform proposals that allow the technology to flourish. In their summary paper the Commission states they believe that NFTs could bring radical legal development, changing how the market, market participants, and the legal system operate and transact with respect to intellectual property rights.¹¹⁴ This is a clear endorsement of NFTs being used for digital copyright protection and gives me reason to believe that in the long term this will not fail due to a lack of reform.

One point to make is that despite being ruled as property in England and Wales, the Law Commission set out in their report that NFTs and digital tokens could constitute a third category of personal property – labelled “data objects.”¹¹⁵ This brings uncertainty over the extent to which NFTs are protected under current law despite their ruling as property, but this is a temporary issue, in my view.

To balance this argument of legitimacy, the fraudulent use of NFTs giving them an appearance of illegitimacy could be credited to the fact that ETH is a public blockchain and anyone can use it. If RAIR vetted all entrants as I have suggested already then, they can offer users some legitimacy in the system – requiring only that they trust RAIR to verify users and their authorship. This is not an ideal solution, as software developers are not legal authorities on authorship or

¹¹³ Law Commission, “Digital Assets” (July 2022) at <https://www.lawcom.gov.uk/project/digital-assets/> accessed 29/08/2022.

¹¹⁴ Law Commission, “Digital Assets: Summary of Consultation Paper” (July 2022) at 1.64.

¹¹⁵ *Ibid* at 1.11(1).

ownership of works, but if evidence of authorship is required for entry, they could develop verification processes to evaluate this. One such solution is to hire an in-house lawyer at first, although I am not convinced they would be open to this.

4.2.

Despite their hype and revenue generated in 2021, there is evidence that interest in NFTs is dropping, rather than increasing.¹¹⁶ There could be several explanations for this, one of these could be the realisation that most NFTs are not connected to the underlying asset in any meaningful sense. YouGov found that the majority of people they studied did not think NFTs represented good value for money, and that a majority are far more likely to view companies offering NFTs negatively than positively.¹¹⁷ This negative perception comes from the current digital art and collectibles uses which are in my view a bad investment, borne out of hopes for a monetary gain by investors who don't seem interested in the underlying art. There are some positive current uses, such as artists selling concert tickets as NFTs, protecting against counterfeit and duplicate tickets¹¹⁸ – but these are a small minority of sales so do not receive the same level of attention.

This generally negative perception of NFTs may prove to be a difficulty as authors will not want to use them as digital copyright protection for fear of it negatively affecting their revenues and public image. For solutions like DDRM to work, a substantial number of popular artists must view NFTs favourably enough to switch from platforms like YouTube or Spotify, to a DDRM model. The consequences of a switch on an author could be a loss in revenues initially as consumers would be forced to pay an extra expense for content to which they may previously have had access.

A final issue of public perception is an inevitable rehash of the DRM debate. This time around the strength of the measures will be even more prohibitive of fair dealing for the purposes of parody, education, or other uses such as remixes of songs. With successful DDRM and copyright protection through a register, the risk of over-licensing and over-enforcement is much greater than previous DRM, as smart contracts will automatically serve notices for takedown, or initiate other proceedings. Likewise, licences will be required for any use of works even where they are not required, encryption tools could also prevent users exploiting works in the ways they want

¹¹⁶ Armstrong, M., "Interest in NFTs has Plummeted" Statista, 11th March 2022 at <https://www.statista.com/chart/27030/google-search-interest-in-nft/> accessed 31/08/2022.

¹¹⁷ Newby, J., "What Do Britons Think of Companies That Offer NFTs?" YouGov 8th March 2022 at <https://business.yougov.com/content/41416-what-do-britons-think-companies-offer-nfts> accessed 31/08/2022.

¹¹⁸ Thomas, L., "NFT Tickets Are the Future of Live Music. Here's Why" NFT Now 28th July 2022 at <https://nftnow.com/features/nft-tickets-are-the-future-of-live-music/> accessed 31/08/2022.

to, even after purchase. These are issues that must be carefully navigated to find adequate solutions, one of such being a questionnaire prior to the use of a work determining if a licence is required,¹¹⁹ although this may turn consumers away from use simply because of added hassle if alternative traditional streaming services are still in use.

4.3.

As I mentioned earlier, a complete removal of intermediaries is problematic and undesirable for the health of the creative industries. Despite being some of the causes for revenue inequality, a lack of transparency, and other issues authors face – digital piracy has always plagued the internet, and it is only because of intermediaries such as Spotify and YouTube that revenues have increased annually. Granted, the intermediaries suspected of the most wrongdoing are not Spotify or YouTube, but ISPs creating high barriers to entering the market and competitively pricing authors out of distributing content on their own.¹²⁰ On this note, if DDRM solutions are provided in the format RAIR is developing, they are simply switching intermediaries, as RAIR can (and should) charge a fee for use of their services. If DDRM is successful, there would be scope for RAIR to increase the fees to use their technology and authors would have incentive to pay these costs because of the unique benefits available.

One example of the potential replacement of intermediaries is the Ethereum Decentralised Autonomous Organisation (DAO) attack. In this case, the DAO is essentially a decentralised fund in which users can borrow a set amount to fund their decentralised Application projects. DAOs can be set up to act as anything. The full details are irrelevant to this point, but an “attacker” found an exploit in the code which allowed them to continuously take money from the DAO, resulting in the fund being drained of 3,641,694 ETH.¹²¹ This would ordinarily be a simple case of theft, but here the ethos of the crypto world is “Code is Law,”¹²² where code is the only and final law – a way of excluding intermediaries (lawyers). So, by their standards they were lawfully robbed of (then) \$50 million, which equates to \$5.6 billion today. The Ethereum developers were forced to fall on their sword and provide the DAO a bailout of sorts by creating a “fork” in the blockchain, which means they started a new chain continuing from the moment the attack started – reversing all transactions made during the attack. The original blockchain containing

¹¹⁹ N39 page 27.

¹²⁰ N44.

¹²¹ Thomson C., ‘The DAO of ETHEREUM: Analyzing the DAO hack, the Blockchain, Smart contracts, and the Law’ (2016) Medium at <<https://medium.com/blockchain-review/the-dao-of-ethereum-e228b93afc79>> accessed 11/07/2022.

¹²² See Quinn, J., “‘Code Is Law’ During The Age Of Blockchain”, Forbes, 17th May 2022 at <<https://www.forbes.com/sites/forbesbusinesscouncil/2022/05/17/code-is-law-during-the-age-of-blockchain/?sh=21124b542adb>> accessed 30/08/2022.

the attack still exists, now called “Ethereum Classic” but the effect of all this is they turned back time. Therefore, code is not law, it can be whatever the developers in control of the chain want it to be. The point here is not that the developers abused their power, they did the right thing in my view, but in doing so they cast severe doubt on the immutability of blockchains, as whilst they cannot change data in blocks, they can start a new chain from where they want to. The success of the fork is also dependent on users of that blockchain opting to use the new version of Ethereum, so in this sense they cannot abuse their power because the network of users will not allow it to happen.

The risk of new intermediaries becoming overbearing is a problem that is specific to DDRM and privatised uses of NFTs, as entry to a copyright blockchain could not be prevented to a legitimate author by the IPO. The downfall of a reliance on companies such as RAIR is as set out above, opening the door to new barriers for entry and ultimately removing the “trustless” aspect of blockchain as now the network’s integrity could be challenged.

4.4.

Blockchain networks had initially been thought to upgrade DRM when it was first introduced which gained large traction, the promise being increased efficiency in licensing and giving greater control to creators. This fell short due to the nature of international copyright – there is no single international “copyright”, but rather a guarantee that whatever a signatory of the Berne Convention deems to be copyright for its own citizens is also granted to works originating from other signing member states. In each work subsists several rights, which is why copyright is referred to as a bundle of rights – meaning that one work under the protection of the Berne Convention has several rights in each of the 176 Member States. Assuming that on average 10 rights fragments subsist in one work, this would mean that one work would have 1760 individual rights to deal with.¹²³ This was a problem for blockchain for several reasons which also apply to NFTs, most of all this fragmentation of rights is difficult to collate and manage as there are limited stores of RMI and even if available this would require huge ongoing cooperation and coordination between on and off-chain sources.¹²⁴ The management of all fragments could be automated through smart contracts in theory by issuing a mass use non-exclusive licence, such as a creative commons licence, which would be valid in every jurisdiction, which leaves them the autonomy to negotiate specific uses under alternative exclusive licences.¹²⁵

¹²³ N81 page 322.

¹²⁴ N81 page 322.

¹²⁵ Ibid.

The downfall of blockchain in managing copyright was also due to the lack of high-quality metadata available due to institutional issues.¹²⁶ This problem still has no solution and would affect Pech's blockchain registry because it would rely on an upload of large stores of copyright information from CMOs. If this proved to be too difficult a problem to solve, I would argue there would still be benefits to starting the register and entering all works with the required data. The current problems would remain with works left off the register, a search for information would be incomplete and so unreliable at first, but the lack of high-quality data should not prevent the initiation of a solution to this problem for future generations.

4.5.

There is also considerable risk involved with an investment in NFTs generally, not least a project aiming to upgrade digital copyright protection. The main risk is the risk of failure to protect against piracy; both of my prospective solutions use blockchain technology and NFTs in a positive manner, this does not prevent pirates from also developing blockchain piracy platforms. Piracy is a constantly developing battle; pirates are not stuck with using the same technology they do today, pirates will upgrade their technology at similar rates to those trying to stop piracy because their business depends on it. This risk of failure depends on the type of work, encrypted music can still be recorded on the average person's smartphone, and websites illegally streaming protected content could develop software that tricks content scanners.¹²⁷ YouTube's Content ID, for example, is subverted constantly, it is also subject to abuse by users falsely claiming authorship. The risk of failure is a significant issue preventing a copyright register, but less so for DDRM because of its smaller scale and lack of reliance on public resources.

The final problem to be faced is the fact that piracy may never be stopped, much like other crimes. This is a practical and ethical question that I cannot cover in this paper but must be mentioned in any discussion of piracy prevention. For piracy to end, all consumers must have access to reasonably priced works, globally, as piracy is an international business serving the needs of people who cannot afford, nor have access to, works available in other countries. For this reason, I believe piracy prevention should be viewed at a domestic level first and foremost – if UK piracy is ended this is clearly an evolution of digital copyright protection.

¹²⁶ N81 page 18.

¹²⁷ Gershgorn, D., "Artificial Intelligence Used by Google to Scan Videos Could Easily Be Tricked by a Picture of Noodles" 4th April 2017, Quartz at <https://qz.com/948870/the-ai-used-by-google-to-scan-videos-could-easily-be-tricked-by-a-picture-of-noodles/> accessed 30/08/2022.

Chapter 5 – Conclusion

In conclusion, NFTs as technology *are* the evolution of digital copyright protection and their use must be attempted at state level with blockchain copyright registries and at business level using developers' models of DDRM.

They can accurately be described as an evolution for the following reasons: they offer authors an alternative way to monetise their works; authors are no longer beholden to intermediaries to protect their works; and finally, they are a total upgrade of DRM technology to decentralised networks, which allows for better encryption and secure storage, offering a solution to the problem of digital piracy.

The uses I have shown NFTs can perform are subject to some major concerns, the biggest of which is regulation, covering the process of minting as a restricted act, and legitimising their use as legal tokens.

Whilst I conclude that they are the evolution of digital copyright protection, this should be taken as the evolution *provided they are used*. I conclude that a copyright register is not likely to be used for several years because of its overall practicality and scale – but this does not mean that if attempted it would evolve how we store RMI, and how we can implement TPM. Until such a time that blockchain registers are commonly used, DDRM can successfully evolve the protection of digital copyright works provided it can legitimately verify an author is the owner of the work.

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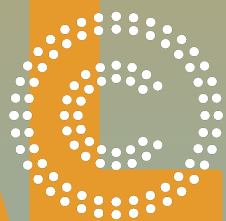
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2022/12 DOI: 10.5281/zenodo.7313051

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